

1. Concept of Standard Normal Distribution

The **Standard Normal Distribution (Z-distribution)** is a special case of the **Normal Distribution** where:

- The mean (μ) = 0
- The standard deviation (σ) = 1
- It follows the **bell-shaped curve** of the normal distribution.

The formula for **Z-score (Standard Score)** is:

$$Z = \frac{X - \mu}{\sigma}$$

where:

- X = data point
- μ = population mean
- σ = standard deviation
- Z = standard score (how many standard deviations X is from μ)

The **Z-score** helps to compare different datasets and find probabilities using the **Z-table**.

2. Properties of Standard Normal Distribution

1. **Symmetrical** around the mean $\mu = 0$.
 2. **Total area under the curve** = 1 (100% probability).
 3. **68-95-99.7 Rule (Empirical Rule)**:
 - 68% of values lie within **1 standard deviation**.
 - 95% of values lie within **2 standard deviations**.
 - 99.7% of values lie within **3 standard deviations**.
 4. The probability of a **single value** is **zero** (as it's a continuous distribution).
 5. Probabilities are found using the **Z-table** (cumulative probability from the left).
-

3. Numerical Problems on Standard Normal Distribution

Example 1: Finding Z-score

Problem:

A class has an average test score of **75** with a standard deviation of **10**. If a student scores **85**, find the Z-score.

Solution:

$$Z = \frac{X - \mu}{\sigma} = \frac{85 - 75}{10} = \frac{10}{10} = 1$$

The student's score is **1 standard deviation above the mean**.

Example 2: Probability of a Score Below a Given Value**Problem:**

In a university entrance exam, scores are normally distributed with $\mu = 500$ and $\sigma = 100$. What is the probability that a randomly selected student scores **less than 400**?

Solution:

$$Z = \frac{X - \mu}{\sigma} = \frac{400 - 500}{100} = \frac{-100}{100} = -1$$

From the **Z-table**, $P(Z < -1) = 0.1587$.

So, the probability that a student scores **less than 400** is **15.87%**.

Example 3: Probability of a Score Above a Given Value**Problem:**

A company's employee salaries follow a normal distribution with a mean of **\$50,000** and a standard deviation of **\$8,000**. What is the probability that a randomly selected employee earns **more than \$60,000**?

Solution:

$$Z = \frac{60000 - 50000}{8000} = \frac{10000}{8000} = 1.25$$

From the **Z-table**, $P(Z < 1.25) = 0.8944$.

To find $P(X > 60000)$:

$$P(X > 60000) = 1 - 0.8944 = 0.1056$$

So, the probability is **10.56%**.

Example 4: Probability Between Two Scores

Problem:
The IQ scores of a population are normally distributed with $\mu = 100$ and $\sigma = 15$. What is the probability that a person has an IQ between 85 and 115?

Solution:
Find Z-scores:

$$Z_1 = \frac{85 - 100}{15} = \frac{-15}{15} = -1$$
$$Z_2 = \frac{115 - 100}{15} = \frac{15}{15} = 1$$

From the Z-table:

$$P(Z < -1) = 0.1587, \quad P(Z < 1) = 0.8413$$
$$P(85 < X < 115) = P(Z < 1) - P(Z < -1) = 0.8413 - 0.1587 = 0.6826$$

So, the probability is **68.26%**.

Example 5: Finding a Score for a Given Probability

Problem:
A university wants to admit students who score in the **top 5%** of a standardized test. The scores follow a normal distribution with $\mu = 500$ and $\sigma = 100$. What is the minimum required score?

Solution:
The **top 5%** means $P(X > ?) = 0.05$, which corresponds to $Z = 1.645$ (from the Z-table).
Now, find X:

$$X = \mu + Z\sigma = 500 + (1.645 \times 100)$$
$$X = 500 + 164.5 = 664.5$$

So, the minimum required score is **665** (rounded up).

4. Summary Table

Scenario	Formula	Example Result
Find Z-score	$Z = \frac{X - \mu}{\sigma}$	$Z = 1$
Find $P(X < a)$	Use Z-table	$P(X < 400) = 15.87\%$

Scenario	Formula	Example Result
Find $P(X > a)$	$1 - P(X < a)$	$P(X > 60,000) = 10.56\%$
Find $P(a < X < b)$	$P(Z < b) - P(Z < a)$	$P(85 < X < 115) = 68.26\%$
Find X for given probability	$X = \mu + Z\sigma$	$X = 665$